

Zaim Feroz

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EDUCATION

University of Waterloo

Candidate for Bachelor of Applied Science, Electrical Engineering

Waterloo, ON

Sep. 2024 – Present

TOOLS & SKILLS

Programming & Firmware

- C++
- C
- Embedded C
- STM32 HAL
- Arduino

Design Tools

- LTSpice
- Altium Designer
- Fusion 360
- STM32CubeIDE
- STM32CubeMX
- PuTTY

Engineering Skills

- Circuit Design
- PCB Design & Layout
- Schematic Capture
- Embedded Systems
- Hardware Prototyping
- GPIO & UART Configuration
- PCB Assembly & Debugging
- 3D CAD Modelling
- Design for Manufacturing (DFM)

PROJECTS

Project 1: Smart Battery Management System (STM32)

Overview

Designed and built a complete **Smart Battery Management System (BMS)** using an **STM32F401RE microcontroller** to monitor battery **voltage, temperature, and current** while protecting a **lithium-ion battery** from unsafe operating conditions. The project integrated **embedded firmware, analog circuit design, PCB development, mechanical enclosure design, and hardware validation** into a **fully functional prototype**. Custom firmware **continuously monitored** battery conditions, controlled **system status LEDs** and the **load MOSFET**, and transmitted **live diagnostic** information through **UART** for debugging and verification.

Objectives

- Design a complete battery management system for a single-cell lithium-ion battery.
- Continuously monitor battery voltage, temperature, and load current.
- Detect overvoltage, undervoltage, overcurrent, and overtemperature conditions.
- Implement warning and fault protection algorithms in Embedded C.
- Control system status LEDs and load MOSFET using an STM32 microcontroller.
- Design a custom two-layer PCB in Altium Designer.
- Validate hardware functionality using breadboard testing before PCB implementation.

Tools Used

- **Firmware Development**
 - Embedded C
 - STM32CubeIDE
 - STM32CubeMX
 - STM32 HAL
 - UART
 - ADC
- **Hardware Design**
 - Altium Designer
 - LTSpice
- **Hardware Validation**
 - Breadboard Prototyping
 - PuTTY
 - STM32 NUCLEO-F401RE

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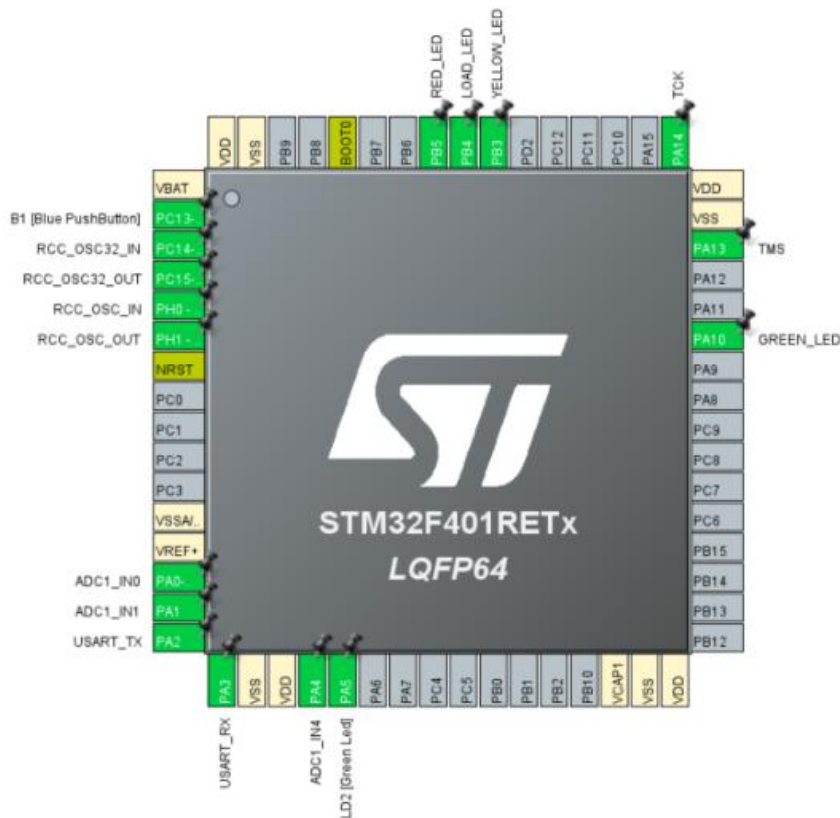
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STM32 Firmware

Developed embedded firmware in Embedded C using STM32CubeIDE to monitor battery voltage, temperature, and current while implementing real-time battery protection algorithms. The firmware continuously sampled analog sensor inputs using the ADC, evaluated battery health against predefined warning and fault thresholds, controlled the system LEDs and load output, and transmitted live diagnostic information through UART.

Firmware Features

- Developed embedded firmware in Embedded C using STM32CubeIDE.
- Configured ADC, UART, and GPIO peripherals using STM32CubeMX.
- Measured battery voltage through a resistor divider.
- Calculated battery temperature using a 10 k Ω NTC thermistor and Beta equation.
- Measured load current using an INA180 current-sense amplifier.
- Implemented overvoltage, undervoltage, overtemperature, and overcurrent protection logic.
- Controlled Green, Yellow, Red, and Blue system status LEDs.
- Implemented real-time UART diagnostics for debugging and system validation.
- Validated firmware using live hardware testing on the STM32 NUCLEO-F401RE.



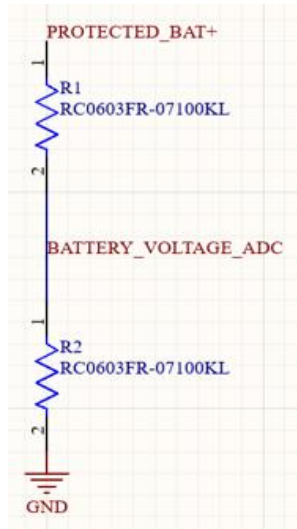
Circuit Design

Designed the complete Smart Battery Management System schematic in Altium Designer using modular hierarchical sheets for improved organization and readability. Individual circuits were designed and validated using LTspice before integration into the final PCB.

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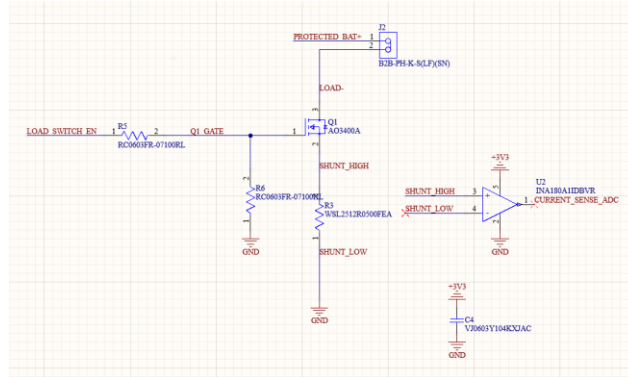
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Voltage Monitoring



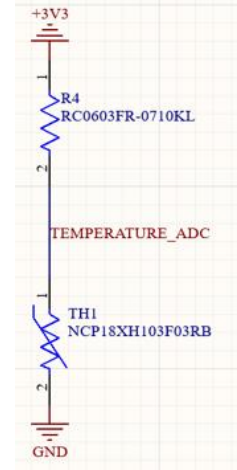
- Designed a resistor divider to safely measure battery voltage.
- Calculated resistor values to maximize ADC resolution.
- Validated operation using LTspice simulation.

Current Monitoring



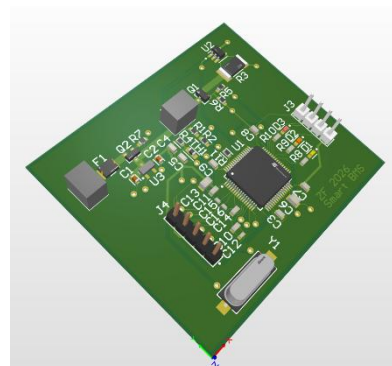
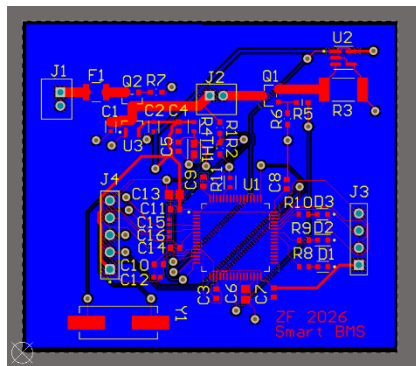
- Designed a current sensing circuit using the INA180 current-sense amplifier.
- Selected a 0.05 Ω shunt resistor for accurate current measurement.
- Verified amplifier output using LTspice simulations.

Temperature Monitoring



- Designed a thermistor voltage divider using a 10 k Ω NTC thermistor.
- Implemented the Beta equation in firmware to calculate battery temperature.
- Verified temperature measurement through hardware testing.

PCB Design

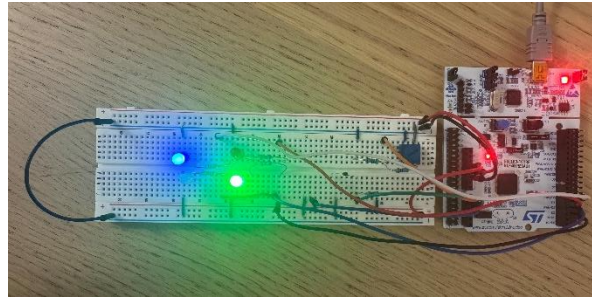


- Designed a compact two-layer PCB in Altium Designer.
- Organized the design into modular schematic sheets.
- Routed power and signal traces using appropriate design rules.
- Added ground pours to improve electrical performance.
- Performed Design Rule Checks (DRC) prior to manufacturing.
- Generated a 3D PCB model to verify component placement and board geometry.

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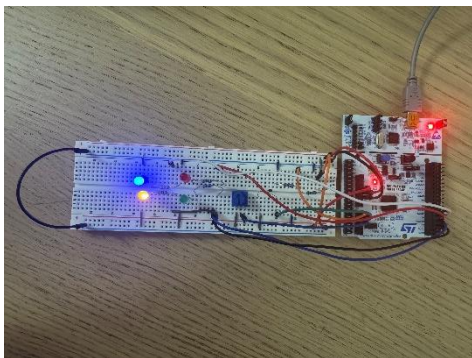
Breadboard Validation



The complete Smart Battery Management System was assembled on a breadboard using an STM32 NUCLEO-F401RE before PCB fabrication. Breadboard testing validated the firmware, analog measurement circuits, and LED control logic prior to hardware manufacturing.

- Verified battery voltage measurements.
- Verified thermistor temperature calculations.
- Verified current measurement calculations.
- Tested UART communication using PuTTY.
- Validated warning and fault detection algorithms.
- Confirmed LED indication logic for all operating states.
- Verified load enable/disable functionality before PCB implementation.

System Validation



Battery (V)	Temp (C)	Current (A)	State	Green	Yellow	Red	Load	Diagnostic Message
3.31	24.2	0.00	NORMAL	ON	OFF	OFF	ON	NORMAL: All measured values are within safe operating limits
3.30	24.0	0.00	NORMAL	ON	OFF	OFF	ON	NORMAL: All measured values are within safe operating limits
3.31	23.9	0.01	NORMAL	ON	OFF	OFF	ON	NORMAL: All measured values are within safe operating limits
3.31	23.9	0.00	NORMAL	ON	OFF	OFF	ON	NORMAL: All measured values are within safe operating limits
3.32	23.9	0.07	NORMAL	ON	OFF	OFF	ON	NORMAL: All measured values are within safe operating limits
3.31	24.1	0.54	NORMAL	ON	OFF	OFF	ON	NORMAL: All measured values are within safe operating limits
3.31	24.2	0.54	NORMAL	ON	OFF	OFF	ON	NORMAL: All measured values are within safe operating limits
3.31	24.2	0.60	NORMAL	ON	OFF	OFF	ON	NORMAL: All measured values are within safe operating limits
3.31	23.9	0.93	NORMAL	ON	OFF	OFF	ON	NORMAL: All measured values are within safe operating limits
3.31	24.0	0.93	NORMAL	ON	OFF	OFF	ON	NORMAL: All measured values are within safe operating limits
3.31	24.3	1.29	NORMAL	ON	OFF	OFF	ON	NORMAL: All measured values are within safe operating limits
3.31	24.3	1.36	NORMAL	ON	OFF	OFF	ON	NORMAL: All measured values are within safe operating limits
3.30	24.1	1.37	NORMAL	ON	OFF	OFF	ON	NORMAL: All measured values are within safe operating limits
3.30	24.1	1.71	WARNING	OFF	ON	OFF	ON	WARNING: HIGH CURRENT FIX: Reduce load before overcurrent shutdown
3.31	24.4	1.76	WARNING	OFF	ON	OFF	ON	WARNING: HIGH CURRENT FIX: Reduce load before overcurrent shutdown
3.30	24.2	1.75	WARNING	OFF	ON	OFF	ON	WARNING: HIGH CURRENT FIX: Reduce load before overcurrent shutdown
3.31	24.2	1.75	WARNING	OFF	ON	OFF	ON	WARNING: HIGH CURRENT FIX: Reduce load before overcurrent shutdown
3.32	24.3	1.75	WARNING	OFF	ON	OFF	ON	WARNING: HIGH CURRENT FIX: Reduce load before overcurrent shutdown
3.31	24.3	1.83	WARNING	OFF	ON	OFF	ON	WARNING: HIGH CURRENT FIX: Reduce load before overcurrent shutdown
3.31	24.4	2.16	FAULT	OFF	OFF	ON	OFF	FAULT: OVERCURRENT FIX: Disconnect load and inspect for overload or short circuit
3.31	24.4	2.47	FAULT	OFF	OFF	ON	OFF	FAULT: OVERCURRENT FIX: Disconnect load and inspect for overload or short circuit
3.31	24.5	2.75	FAULT	OFF	OFF	ON	OFF	FAULT: OVERCURRENT FIX: Disconnect load and inspect for overload or short circuit
3.31	24.6	3.30	FAULT	OFF	OFF	ON	OFF	FAULT: OVERCURRENT FIX: Disconnect load and inspect for overload or short circuit
3.30	24.5	3.30	FAULT	OFF	OFF	ON	OFF	FAULT: OVERCURRENT FIX: Disconnect load and inspect for overload or short circuit
3.31	24.6	3.30	FAULT	OFF	OFF	ON	OFF	FAULT: OVERCURRENT FIX: Disconnect load and inspect for overload or short circuit
3.31	24.6	3.30	FAULT	OFF	OFF	ON	OFF	FAULT: OVERCURRENT FIX: Disconnect load and inspect for overload or short circuit
3.30	24.4	3.30	FAULT	OFF	OFF	ON	OFF	FAULT: OVERCURRENT FIX: Disconnect load and inspect for overload or short circuit

Comprehensive system testing was performed by simulating battery operating conditions using potentiometers and a thermistor. Battery voltage, temperature, and current were varied across their operating ranges to verify that the firmware correctly transitioned between Normal, Warning, and Fault states while updating the LEDs and UART diagnostic output.

- Verified undervoltage warning and fault detection.
- Verified overvoltage warning and fault detection.
- Verified overcurrent warning and fault detection.
- Verified temperature monitoring functionality.
- Confirmed correct LED operation for all system states.
- Validated real-time UART diagnostic reporting.

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Project 2: STM32 Password Lock System

Overview

Designed and built a complete **embedded password lock system** using an **STM32 microcontroller**, integrating custom firmware, PCB design, mechanical enclosure design, and hardware prototyping into a **fully functional device**. The project began with **circuit prototyping** on a breadboard to **validate functionality** before progressing to schematic capture and PCB layout in **Altium Designer**. A **custom enclosure** was then designed in **Fusion 360** to securely house the electronics while providing access to the user interface through a transparent lid, recessed button panel, and integrated buzzer vent. The project provided experience with the **complete embedded systems development workflow**, from hardware prototyping and firmware development to PCB design, mechanical integration, and final assembly.

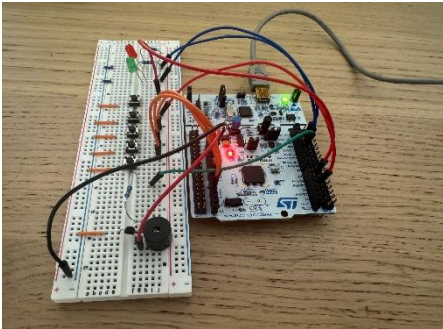
Objectives

- Design and prototype a complete embedded password lock system.
- Validate hardware functionality using a breadboard prototype before PCB design.
- Develop custom firmware using an STM32 microcontroller.
- Design a professional manufacturable PCB in Altium Designer.
- Integrate electrical and mechanical designs into a single product.
- Design a custom enclosure to securely house the electronics.

Tools Used

- **Firmware Development**
 - STM32CubeIDE
 - STM32Cube MX
- **Hardware Prototyping**
 - Breadboard Prototyping
 - PuTTY
- **PCB Design**
 - Altium Designer
- **CAD**
 - Autodesk Fusion 360

Breadboard Prototype



- Built and tested the complete circuit on a breadboard before PCB development.
- Verified the operation of push buttons, LEDs, buzzer, and UART communication.
- Identified and corrected wiring and firmware issues prior to PCB design.
- Confirmed overall system functionality before transitioning to a custom PCB.

STM32 Firmware

Developed embedded firmware in **Embedded C** using **STM32CubeIDE** to control a **password-protected lock system**. The firmware interfaces with multiple hardware peripherals and manages user authentication, feedback, and debugging.

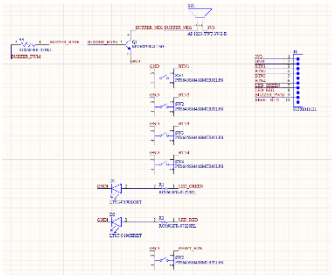
- Developed embedded firmware in Embedded C using STM32CubeIDE.
- Configured **STM32 peripherals** and **GPIO pin assignments** using **STM32CubeMX**.
- Implemented **password verification** and **user authentication logic**.
- Processed push-button inputs and stored **user-entered password sequences**.
- Controlled LEDs and a buzzer to provide **visual and audible user feedback**.
- Added a **password change mode** with **verification** of the existing password.
- Implemented a **three-attempt security lockout** with a **timed cooldown period**.
- Utilized **UART communication** with **PuTTY** for real-time debugging and system monitoring.
- Integrated firmware with custom hardware designed in Altium Designer.

Complete firmware source code, STM32CubeMX pin configuration, and project files are available on [GitHub](#).

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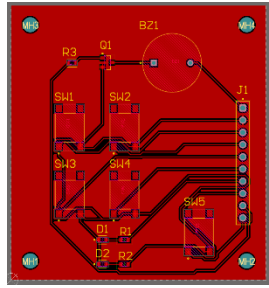
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Altium Design Schematic



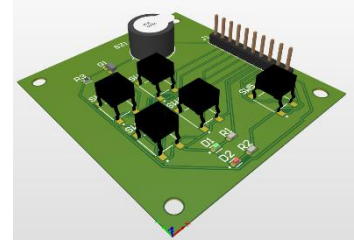
- Designed the complete embedded system schematic in Altium Designer.
- Integrated the STM32 microcontroller, push buttons, LEDs, buzzer, and programming interface.
- Managed component libraries, net labels, and electrical connectivity.
- Prepared the design for PCB layout and manufacturing.

PCB Layout



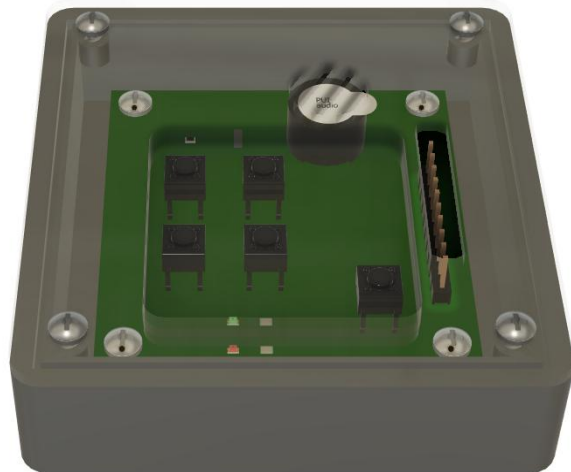
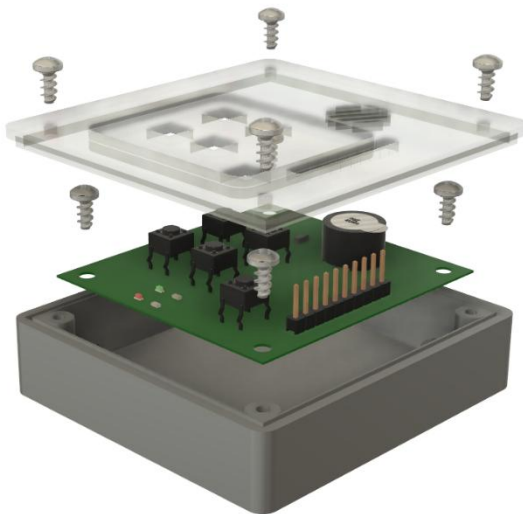
- Designed a compact PCB layout with organized component placement.
- Routed signal and power traces while maintaining clean electrical connectivity.
- Added mounting holes and connectors to support enclosure integration.
- Optimized the layout for manufacturability and assembly.

PCB 3D View



- Generated a 3D model to verify component placement and board geometry.
- Confirmed compatibility between the PCB and enclosure before manufacturing.
- Used the 3D model to assist with mechanical enclosure design.

Fusion 360 Design



- Designed a custom enclosure specifically around the **PCB dimensions**.
- Added **PCB mounting standoffs** for secure board installation.
- Designed integrated **lid screw bosses** for reliable enclosure assembly.
- Implemented a **locating lip and matching groove** for accurate lid alignment.
- Created a **recessed button section** to improve button accessibility and tactile feedback.
- Designed a **slotted buzzer vent** to improve sound transmission while protecting internal components.
- Added **connector cutouts** for external access to the PCB.
- **Imported 3D hardware** models to verify screw fitment and assembly.
- Validated the complete mechanical assembly using an exploded view before finalizing the design.
- Applied appropriate wall thicknesses, manufacturing clearances, and rounded corners for a professional and 3D-printable enclosure.

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Project 3: 12V to 5V Buck Converter

Overview

Designed a **12V to 5V DC-DC buck converter** to efficiently **step-down voltage** using switching power electronics. The project involved **circuit simulation in LTSpice**, schematic capture and **PCB layout in Altium Designer**, and **mechanical enclosure design in Fusion 360**. The goal was to gain experience with the complete **hardware design workflow**, from simulation to a **manufacturable PCB** and finished enclosure.

Objectives

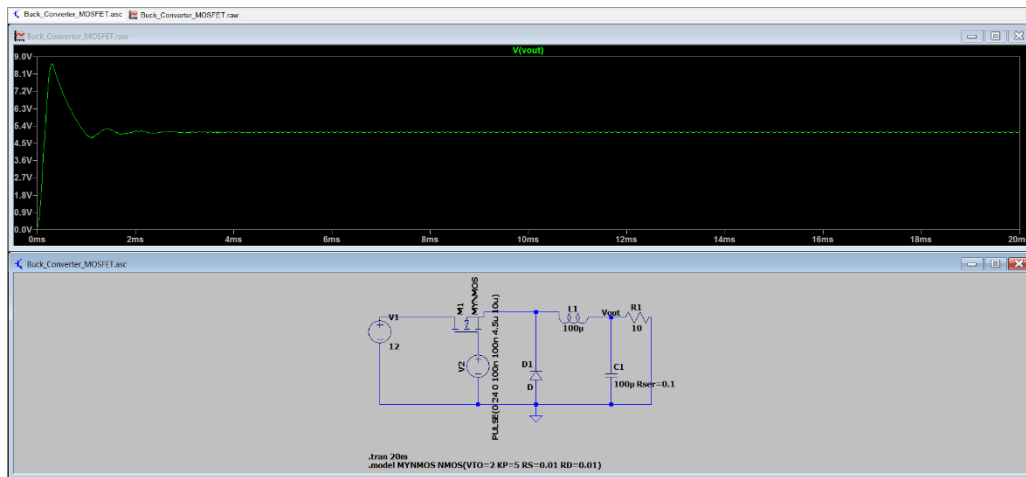
- Understand the operating principles of a buck converter
- Simulate switching behavior and analyze output characteristics
- Design a complete schematic and PCB in Altium Designer
- Create a mechanical enclosure in Fusion 360
- Gain experience with end-to-end hardware design

Tools Used

- **Simulation**
 - LTSpice
- **PCB Design**
 - Altium Designer
- **CAD**
 - Autodesk Fusion 360

LTSpice Simulation

Schematic

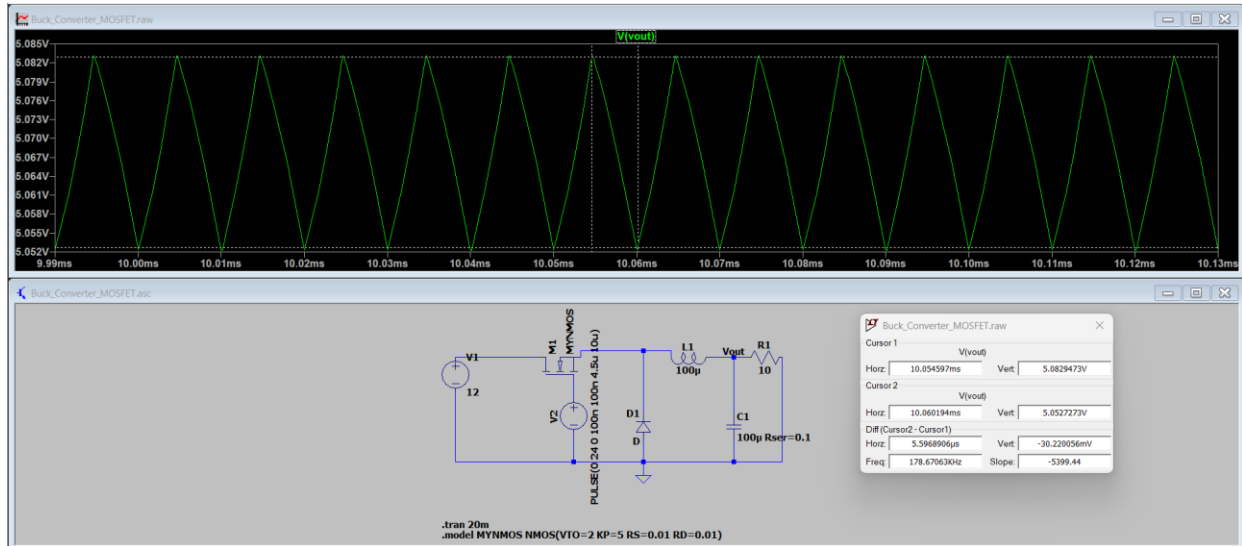


- Designed a 12V to 5V buck converter using an NMOS transistor, Schottky diode, 100 µH inductor, and 100 µF output capacitor.
- Selected component values to achieve a stable 5V output at a nominal load current of 0.5A.
- Simulated switching behavior and analyzed voltage regulation before moving to PCB design.
- Verified circuit operation and observed energy transfer through the inductor and output filter.

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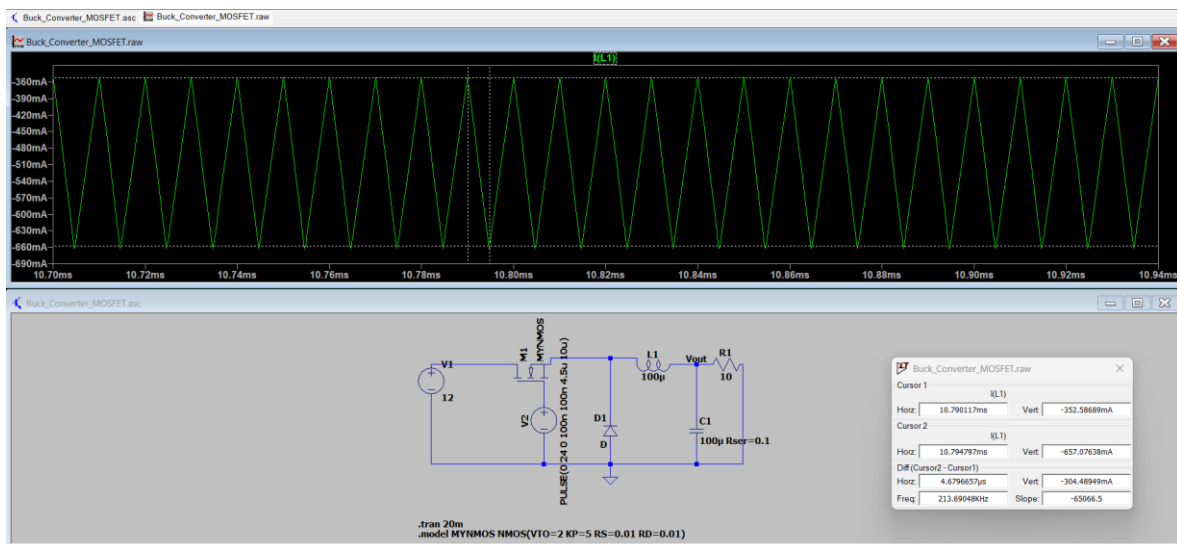
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Output Voltage Waveform



- Measured the output voltage to verify regulation from a 12V input to a stable 5V output.
- Analyzed output ripple and confirmed the effectiveness of the 100 μF output capacitor.
- Verified operation at a 10 Ω load resistance, corresponding to an output current of 0.5A.
- Observed how switching frequency and component selection affect voltage stability.

Inductor Current Waveform



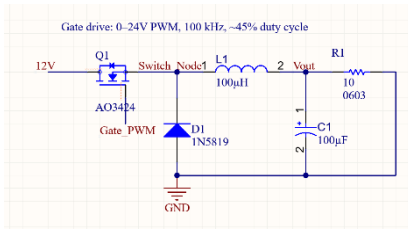
- Observed the inductor charging and discharging during each switching cycle.
- Measured inductor current ripple to verify proper buck converter operation.
- Analyzed energy transfer from the switching stage to the output load.
- Verified operation at a nominal output current of 0.5A with a 10 Ω load.

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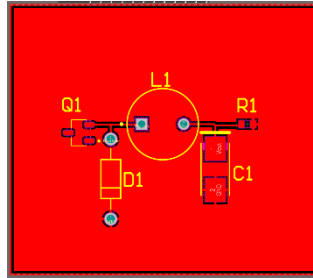
Altium Design

Schematic



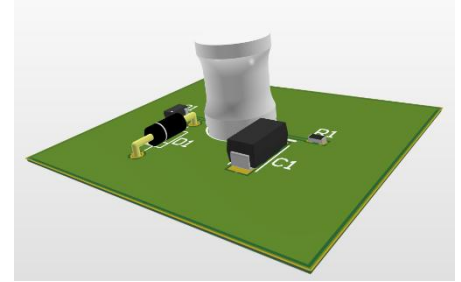
- Recreated the LTSpice design in Altium Designer.
- Selected component footprints and organized schematic connections.
- Learned how to work with component libraries and net labels.
- Prepared the design for PCB layout and manufacturing.

PCB Layout



- Designed a compact PCB layout with optimized component placement.
- Implemented a ground plane to improve current return paths and reduce noise.
- Routed the switching node and high-current traces carefully to improve performance.
- Gained experience transitioning from schematic to a physical PCB design.

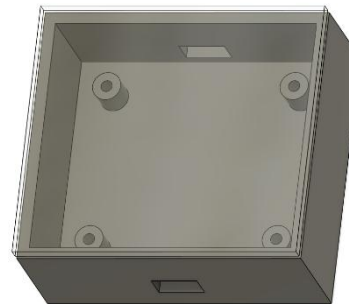
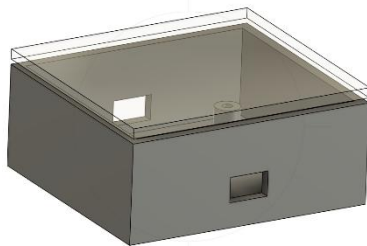
PCB 3D View



- Generated a 3D PCB model to verify component placement and board geometry.
- Visualized the physical arrangement of components before manufacturing.
- Ensured the PCB dimensions were compatible with the enclosure design.
- Explored the relationship between electrical and mechanical design.

Fusion 360 Design

CAD Enclosure



- Designed a custom enclosure in Fusion 360 to house the PCB.
- Added PCB mounting standoffs, screw holes and input/output cutouts.
- Created a transparent lid to showcase the internal electronics.
- Learned mechanical design principles including clearances, wall thickness, and part assembly.